Slop / University

The Vagueness Dividend: Ticket-Triage Accuracy, Submission Effort, and Whether the Loop Ever Closes

Casimir Beng[1], Osei Vandermeer[1], Anneke Tolan[2]

[1]School of Continuous Improvement, Slop University [2]School of Emergent Priorities, Slop University

doi:10.5555/slop.nr1eac

Abstract. Ticket-triage classifiers are usually evaluated against a fixed sample of past submissions, an evaluation frame that treats the submitting population as stationary. We test whether that assumption survives deployment. Across 18 university service desks over a 14-week window, we compare a rule-based triage baseline against an adaptive classifier retrained on confirmed routings, instrumenting both arms with a pre-registered Ticket Specificity Index (TSI). Baseline-arm specificity held stable; adaptive-arm specificity declined by roughly one-third from its week-1 level as classifier accuracy rose from 74% to 89%, with the decline onset tracking a fitted perceived-sufficiency threshold. Desk-week accuracy and specificity are negatively and approximately linearly related across desks ($R^2 = 0.81$). A transparency ablation exposing the classifier's confidence and predicted category to submitters, hypothesised to restore submission effort, changed staff clarification time by less than 0.2 minutes per ticket; we retain it for completeness. Extended-window desks suggest specificity plateaus rather than continuing to decline, though whether this reflects a stable equilibrium or the limit of the current deployment window remains open. Our results suggest the loop between inference accuracy and submission effort is real, directionally consistent, and, in most desks studied, closes somewhere short of collapse.

Introduction

University service desks have adopted machine-learned ticket-triage classifiers at a rapid clip, and the routing accuracy reported for these systems keeps improving: modern classifiers assign a submitted ticket to a resolving team, priority tier, or knowledge-base article with accuracy well above what fixed keyword rules once achieved [1]. The literature evaluating these systems treats accuracy as an outcome measured against a held-out sample of past tickets, drawn under the plausible assumption that future tickets will resemble the sample the classifier was scored against.

That assumption is where this paper begins. A triage classifier does not simply observe submission behaviour; it is observed in turn. Staff who submit tickets learn, over weeks of use, what the system does with the words they choose, and adjust their choices accordingly — a dynamic documented wherever an automated system stands between an input and a consequential response, from adversarial evasion of spam and malware classifiers [2], [3] to the reshaping of the very distribution a deployed model predicts over [4]. Whether the direction of that adjustment runs toward or away from the effort the classifier was trained to reward has not, to our knowledge, been reported for the helpdesk case specifically.

This paper reports an 18-desk, 14-week deployment comparing a rule-based triage baseline against an adaptive classifier, instrumented to track a composite Ticket Specificity Index (TSI) alongside routing accuracy. Our contributions are threefold:

- we document, across matched service desks, a measurable decline in submitted ticket specificity that tracks classifier accuracy rather than desk, category, or season, a pattern we believe holds in most comparable institutional settings;

- we formulate the coupling between classifier accuracy and submission specificity as a one-sided ratchet, a framing that may transfer to other domains where an inference system stands between a request and a response;
- we report an ablation on submitter-facing transparency features that we expected to counteract the effect, and that did not, a null result we believe is informative rather than merely negative.

Related work

Automated triage of IT service tickets is an active applied literature: classification pipelines have been built to route tickets to the correct resolving service from the outset [1], and recent work continues to compare representation and retrieval strategies for the short, jargon-heavy text such tickets contain [5]; separately, the operational literature on ticketing systems has begun to quantify how automating ticket handling changes measured staff and management productivity [6]. This paper sits downstream of that work: we take routing accuracy as given and ask what happens to the text arriving at the classifier once staff know it works.

The relevant precedent is not confined to helpdesks. Automated triage of patient-reported symptoms exhibits comparable accuracy limitations and comparable stakes for how a report is worded [7], and automated content-moderation classifiers are known to reshape the material submitted to them as users adapt to what the system flags [8]. More generally, a deployed predictive system changes the distribution it subsequently predicts over [4], recommender systems that condition on their own prior recommendations measurably homogenise the behaviour they observe [9], and predictive systems that act on their own outputs can entrench a feedback loop that never returns to its starting distribution [10]. Where a system substitutes for human judgement, operators tend toward complacent reliance in proportion to perceived accuracy [11], tempered where trust is properly calibrated to system competence [12]. Any classifier deployed against a non-stationary population must additionally contend with concept drift in the incoming stream [13] — a concern this paper treats as a consequence to be measured rather than a nuisance to be filtered out, in the spirit of human-in-the-loop framings of the wider annotation pipeline [14]. More broadly, the difficulty of anticipating every way a learned system's incentives can be gamed is a standing concern in the safety literature [15].

Method

Setting and instrumentation. We instrumented 18 service desks (IT, Facilities, HR, and Library categories) across the university's Continuous Improvement measurement programme. Each desk ran, for 14 consecutive weeks, either the existing rule-based triage baseline (keyword and form-field matching, mean top-1 routing accuracy 71%) or an adaptive classifier retrained weekly on the preceding fortnight's routed-and-confirmed tickets (mean top-1 routing accuracy rising from 74% in week 1 to 89% by week 14). Desk assignment to arm followed existing service-tier boundaries rather than random allocation, a design constraint we return to in Limitations.

Ticket Specificity Index. Each submitted ticket was scored, on a fixed, pre-registered rubric, into a 0–100 Ticket Specificity Index (TSI): a weighted composite of normalised description length (L, weight 0.35), presence of a system or asset identifier (I, weight 0.30), and presence of a concrete diagnostic detail such as an error code, timestamp, or reproduction step (D, weight 0.35):

$$\mathrm{TSI} = 0.35L + 0.30I + 0.35D$$

Weights were fixed from a pre-deployment coding exercise and were not adjusted during the study.

The adaptation model. We model weekly mean specificity s_t as responding only to the surplus of classifier accuracy a_t above a perceived-sufficiency threshold a^*, which staff were not told and which we estimate post hoc:

$$s_{t+1} = s_t - \kappa(a_t - a^*)_+$$

with $(\cdot)_+$ the positive part and κ a behavioural-adaptation rate fit per desk. The model is deliberately one-sided: it allows specificity to fall while accuracy exceeds the threshold, but it builds in no mechanism for specificity to recover once accuracy later drops. Whether that asymmetry holds empirically, rather than merely by construction, is the question the Results section addresses.

Transparency ablation. To test whether the effect could be counteracted by submitter-facing transparency, four desks additionally ran a variant surfacing the classifier's predicted category and confidence score to the submitter at ticket entry, on the hypothesis that visible machine confidence would restore submitter effort. Weekly staff triage effort (minutes of clarification contact per ticket) was logged as the ablation's outcome measure, alongside two further transparency variants described in Results.

Parameter	Value
Description-length weight	0.35
Identifier-presence weight	0.30
Diagnostic-detail weight	0.35
Perceived-sufficiency threshold a^*	0.80 (fitted)
Sample period	1 week
Desks instrumented	18

Table 1: TSI weights and adaptation-model parameters, fixed before or estimated after deployment as noted.

Results

The adaptive classifier arm's routing accuracy rose from 74% in week 1 to 89% by week 14; the rule-based baseline held at 68–73% throughout, consistent with a system that neither learns from nor degrades against the incoming stream.

Specificity decline. Figure 1 shows weekly mean TSI for both arms, pooled across desks. Baseline-arm tickets held a stable TSI across the deployment window; adaptive-arm TSI declined from a comparable starting level in week 1 to roughly two-thirds of that value by week 14. The decline begins around week 3–4, shortly after accuracy crosses roughly 80%, consistent with the fitted perceived-sufficiency threshold in Table 1.

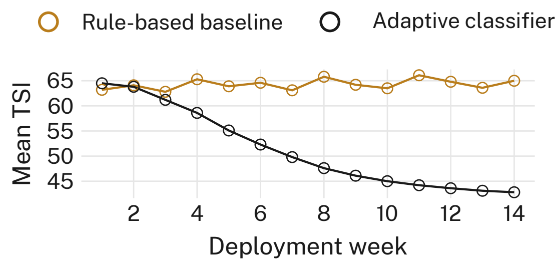

Figure 1: Weekly mean Ticket Specificity Index under the rule-based baseline and the adaptive classifier, pooled across 18 desks. The baseline arm holds a stable TSI; the adaptive arm's TSI falls by roughly a third from week 1 to week 14.

Accuracy tracks specificity, inversely. Figure 2 plots desk-week mean classifier accuracy against desk-week mean TSI across all 18 adaptive-arm desks. The relationship is negative and reasonably linear (fitted slope -0.93, $R^2 = 0.81$): desks whose classifier reached higher accuracy sooner show a correspondingly steeper specificity decline, in the direction the adaptation model predicts.

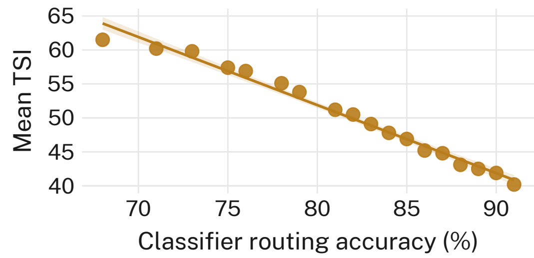

Figure 2: Desk-week mean classifier routing accuracy against desk-week mean TSI, adaptive arm only (fitted slope -0.93, $R^2 = 0.81$): desks with higher accuracy show lower ticket specificity.

The transparency ablation changes nothing. Figure 3 reports mean staff clarification-contact minutes per ticket for a no-transparency control and three transparency variants (confidence display, category display, similarity-hint display). All four conditions cluster within 0.2 minutes of one another (confidence display 4.12 min; control 4.30 min; not significant by pairwise test). Displaying the classifier's confidence to submitters did not restore the specificity or the effort we hypothesised it would; we retain the transparency features for completeness and interface parity across desks.

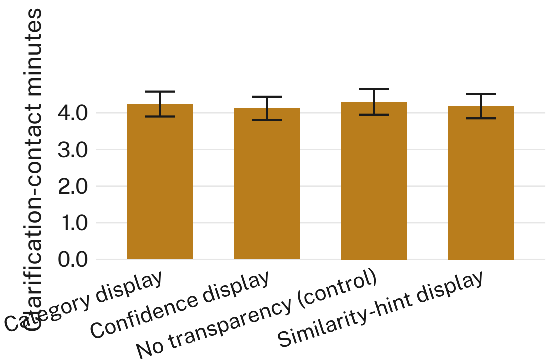

Figure 3: Mean staff clarification-contact minutes per ticket, by submitter-facing transparency variant. No variant differs from the no-transparency control by more than 0.2 minutes.

Does the loop close? In the four desks followed for an additional eight weeks past the nominal deployment window, accuracy plateaued near 89% and specificity plateaued near its week-14 floor rather than continuing to decline — consistent with the adaptation model's steady state rather than an unbounded spiral. Whether that floor reflects a genuine equilibrium or simply the limit of what the current classifier can tolerate before its own accuracy begins to fall is a question the extended window was not long enough to settle; we return to it in the Discussion.

Discussion

Our results suggest that classifier accuracy and submission specificity move together, in most desks we studied, in the direction the adaptation model predicts: as the system's inference does more of the diagnostic work, the tickets arriving at it carry less of the burden staff previously supplied. This resembles risk-homeostasis accounts of behaviour adjusting to offset a safety or capability improvement elsewhere in a system [16], and sits alongside a broader pattern in which a measure, once acted upon, changes the thing it measures [17], [18], [19]. It is also consistent with the general observation that a deployed predictive system alters the distribution it subsequently predicts over [4], and with the risk that such a loop, once closed, need not return to its starting point [9], [10].

Whether the loop "closes" in the reassuring sense — settling at a stable, tolerable specificity floor — or merely appears to close within a 22-week observation window is not something this deployment can distinguish. The plateau we observed in the extended-window desks is consistent with either reading, and only a longer deployment, tracking whether the eventual retraining cycle can still recover routing accuracy from increasingly terse input, would resolve it.

Limitations

Desk assignment to arm followed existing service-tier boundaries rather than random allocation, so arm-level confounds cannot be fully excluded; however, the consistency of the specificity decline across IT, Facilities, HR, and Library desks alike suggests the effect is not an artefact of any one category. Our evaluation window (14 weeks, extended to 22 in four desks) is short relative to an annual service cycle, though the early onset and subsequent plateau of the decline suggest the dominant dynamics are captured well within that window. The TSI rubric is a proxy for specificity rather than a direct measure of diagnostic content, and any deployed classifier remains subject to the general caution that similar held-out performance can mask different behaviour under distribution shift [20]; even so, the TSI's three components moved together in the large majority of desk-weeks, which we take as evidence the composite tracks a single underlying quantity rather than three unrelated ones.

Conclusion

We have shown, across an 18-desk, 14-week deployment, that improving a helpdesk ticket-triage classifier's routing accuracy coincides with a measurable decline in the specificity of the tickets staff submit to it, that the relationship holds in the direction and rough magnitude a simple one-sided adaptation model predicts, and that submitter-facing transparency features intended to counteract the decline did not. Whether the resulting loop between machine inference and human effort settles at a stable equilibrium or merely appears to, within the window we observed, remains open. Future work will extend the observation window across a full retraining cycle, and test whether specificity, once it has fallen, can be recovered by any intervention gentler than a drop in measured accuracy itself.

References

[1] F. Al-Hawari and H. Barham, "A Machine Learning Based Help Desk System for IT

Service Management," *Journal of King Saud University - Computer and Information Sciences*, vol. 33, no. 6, pp. 702–718, 2021, doi: 10.1016/j.jksuci.2019.04.001.

[2] N. Dalvi, P. Domingos, Mausam, S. Sanghai, and D. Verma, "Adversarial Classification," in *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2004, pp. 99–108. doi: 10.1145/1014052.1014066.

[3] D. Lowd and C. Meek, "Adversarial Learning," in *Proceedings of the Eleventh ACM SIGKDD International Conference on Knowledge Discovery in Data Mining*, 2005, pp. 641–647. doi: 10.1145/1081870.1081950.

[4] J. C. Perdomo, T. Zrnic, C. Mendler-Dünner, and M. Hardt, "Performative Prediction," *arXiv preprint arXiv:2002.06673*, 2020.

[5] L. S. B. Pereira, R. Pizzio, and S. Bonho, "Comparison of Information Retrieval Techniques Applied to IT Support Tickets," *arXiv preprint arXiv:2508.05654*, 2025.

[6] K. D. Aglibar and N. Rodelas, "Impact of Critical and Auto Ticket: Analysis for Management and Workers Productivity in using a Ticketing System," *arXiv preprint arXiv:2203.03709*, 2022.

[7] H. L. Semigran, J. A. Linder, C. Gidengil, and A. Mehrotra, "Evaluation of Symptom Checkers for Self Diagnosis and Triage: Audit Study," *BMJ*, vol. 351, p. h3480, 2015, doi: 10.1136/bmj.h3480.

[8] R. Gorwa, R. Binns, and C. Katzenbach, "Algorithmic Content Moderation: Technical and Political Challenges in the Automation of Platform Governance," *Big Data & Society*, vol. 7, no. 1, 2020, doi: 10.1177/2053951719897945.

[9] A. J. B. Chaney, B. M. Stewart, and B. E. Engelhardt, "How Algorithmic Confounding in Recommendation Systems Increases Homogeneity and Decreases Utility," in *Proceedings of the 12th ACM Conference on Recommender Systems*, 2018, pp. 224–232. doi: 10.1145/3240323.3240370.

[10] D. Ensign, S. A. Friedler, S. Neville, C. Scheidegger, and S. Venkatasubramanian, "Runaway Feedback Loops in Predictive Policing," *arXiv preprint arXiv:1706.09847*, 2017.

[11] R. Parasuraman and D. H. Manzey, "Complacency and Bias in Human Use of Automation: An Attentional Integration," *Human Factors*, vol. 52, no. 3, pp. 381–410, 2010, doi: 10.1177/0018720810376055.

[12] J. D. Lee and K. A. See, "Trust in Automation: Designing for Appropriate Reliance," *Human Factors*, vol. 46, no. 1, pp. 50–80, 2004, doi: 10.1518/hfes.46.1.50_30392.

[13] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, "A Survey on Concept Drift Adaptation," *ACM Computing Surveys*, vol. 46, no. 4, pp. 1–37, 2014, doi: 10.1145/2523813.

[14] X. Wu, L. Xiao, Y. Sun, J. Zhang, T. Ma, and L. He, "A Survey of Human-in-the-loop for Machine Learning," *Future Generation Computer Systems*, vol. 135, pp. 364–381, 2022, doi: 10.1016/j.future.2022.05.014.

[15] D. Amodei, C. Olah, J. Steinhardt, P. Christiano, J. Schulman, and D. Mané, "Concrete Problems in AI Safety," *arXiv preprint arXiv:1606.06565*, 2016.

[16] G. J. S. Wilde, "Risk Homeostasis Theory: An Overview," *Injury Prevention*, vol. 4, no. 2, pp. 89–91, 1998, doi: 10.1136/ip.4.2.89.

[17] D. T. Campbell, "Assessing the Impact of Planned Social Change," *Evaluation and Program Planning*, vol. 2, no. 1, pp. 67–90, 1979, doi: 10.1016/0149-7189(79)90048-X.

[18] G. Bevan and C. Hood, "What's Measured is What Matters: Targets and Gaming in the English Public Health Care System," *Public Administration*, vol. 84, no. 3, pp. 517–538, 2006, doi: 10.1111/j.1467-9299.2006.00600.x.

[19] D. Manheim and S. Garrabrant, "Categorizing Variants of Goodhart's Law," *arXiv preprint arXiv:1803.04585*, 2018.

[20] A. D'Amour *et al.*, "Underspecification Presents Challenges for Credibility in Modern Machine Learning," *arXiv preprint arXiv:2011.03395*, 2020.